Requirements Table-*Micro-Lending & Peer Credit Risk Assessor*

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SEC: 'C'

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute borrower credit risk ratings based on debt-to-income ratio, repayment history, and employment stability indicators.	High	Risk tier (Low/Med/High) is accurately assigned for a complete financial profile. Fail: An incomplete financial profile results in a credit approval instead of being rejected/flagged.	Core risk-classification function; protects lenders from unassessed default exposure. (Given)
FR-002	Functional	The system shall allow a borrrower to submit a loan request specifying principal amount, purpose, and repayment tenure.	High	A request with amount within the platform cap and tenure between 1-36 months is stored with status "Pending Assessment" and a unique Loan Request ID within 3 seconds. Fail: A request missing amount, purpose, or tenure is accepted.	The validated loan request is the entry point that triggers risk assessment, scheduling, and lender matching.
FR-003	Functional	Upon a peer lender funding a loan request, the system shall automatically generate a structured EMI repayment schedule showing due dates, principal, interest, and outstanding balance per installment.	High	For a funded loan, the schedule row count equals the agreed tenure in months and the sum of principal components equals the funded amount (+/- Rs.1 rounding). Fail: The schedule total mismatches the funded amount or a due date is missing.	An audit able schedule lets both parties track obligations and lets the system detect missed EMIs.
FR-004	Functional	The system shall automatically trigger a default warning notification to the peer lender and borrower when a scheduled EMI is not received within 3 days of its due date.	Medium	A warning (in-app + email) is logged and dispatched within 1 hour of the 3-day grace period lapsing. Fail: No warning is sent after 5+ days of non-payment.	Early automated default detection protects lender capital and enables timely recovery action.
FR-005	Functional	The system shall allow a peer lender to browse and filter available loan listings by risk tier, requested amount, and tenure.	Medium	Applying a filter (e.g., "Low risk", <= Rs.50,000) returns only listings matching all selected criteria within 2 seconds. Fail: Filtered results include listings outside the specified criteria.	Enables lenders to make informed, risk-adjusted funding decisions and improves matching efficiency.
NFR-001	Nonfunctional (Performance & Security)	Loan repayment transactions and EMI distribution must execute inside database ACID transactions to prevent double-spending.	High	Bench marking tests confirm target latency and security standards under simulated peak load. Fail: A simulated concurrent-write test produces a duplicated or lost EMI record.	Financial correctness under concurrent access is non-negotiable for a platform handling real money movement. (Given)
NFR-002	Nonfunctional (Security)	The system shall encrypt all borrower personal and financial data at rest (AES-256) and in transit (TLS 1.2+), consistent with applicable data-protection regulations.	High	A security scan confirms no financial/PII field is stored or transmitted encrypted. Fail: Any PII/financial field is found in plain text in storage or in a network capture.	Borrower financial data is highly sensitive; encryption is required to maintain user trust and regulatory compliance.

FR-001 and NFR-001 are the instructor-supplied baseline requirements; FR-002 through FR-005 and NFR-002 were authored to complete the set to exactly five FRs and two NFRs as required by the deliverable spec.